Question-10.7.71

EE25BTECH11022 - sankeerthan

Question

The area formed by the positive X axis and the normal and the tangent to the circle$x^2 + y^2 = 4$ at $\left(1, \sqrt{3}\right)$ is ______.

Solution

Equation of the circle

$$\mathbf{X}^T\mathbf{X} = 4, \text{where}\mathbf{X} = \begin{pmatrix} x \\ y \end{pmatrix} \tag{1}$$

Given point of contact

$$\mathbf{P} = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \tag{2}$$

Equation of the tangent at **P**

$$\mathbf{P}^T\mathbf{X} = 4 \tag{3}$$

Solution

Let intersection of tangent with X-axis be $\mathbf{A} = \begin{pmatrix} a \\ 0 \end{pmatrix}$

$$\mathbf{P}^T\mathbf{A} = 4 \tag{4}$$

$$\begin{pmatrix} 1 & \sqrt{3} \end{pmatrix} \begin{pmatrix} a \\ 0 \end{pmatrix} = 4 \implies a = 4 \tag{5}$$

$$\mathbf{A} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \tag{6}$$

Equation of the normal at $\mathbf{P}$

$$\mathbf{X} = t \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}, t \in \mathbb{R} \tag{7}$$

Let intersection of normal with X-axis be $\mathbf{N} = \begin{pmatrix} x \\ 0 \end{pmatrix}$

$$\mathbf{N} = t \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} 0 = t\sqrt{3} \implies t = 0 \implies \mathbf{N} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{8}$$

Area of triangle formed by points **O**,**A**,**P**

$$\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \tag{9}$$

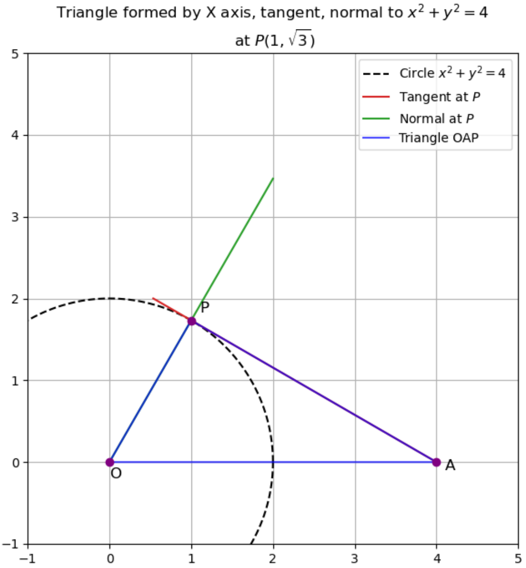
$$S = \frac{1}{2} \left| \begin{pmatrix} 0 & 0 & 1 \\ 4 & 0 & 1 \\ 1 & \sqrt{3} & 1 \end{pmatrix} \right| \tag{10}$$

$$= \frac{1}{2} |0 \left(0 - \sqrt{3}\right) - 0 \left(4 - 1\right) + 1 \left(4\sqrt{3} - 0\right) | \tag{11}$$

$$= \frac{1}{2} |4\sqrt{3}| = 2\sqrt{3} \tag{12}$$

Therefore, required area formed $= 2\sqrt{3}$

graph



Triangle formed by X axis, tangent, normal to $x^2 + y^2 = 4$ at $P(1, \sqrt{3})$
Circle $x^2 + y^2 = 4$
Tangent at P
Normal at P
Triangle OAP
P
O
A
5
4
3
2
1
0
−1
−1
0
1
2
3
4
5

C-Code

```c
// code.c
#include <stdio.h>
#include <math.h>

// Output: pts[9] = [O_x, O_y, A_x, A_y, P_x, P_y, T_x, T_y, N_x,
    N_y]
// O: origin, A: tangent-X, P: point, T: tangent(another), N:
    normal(another)
void get_points(double* pts) {
    // O = (0, 0)
    pts[0] = 0.0; pts[1] = 0.0;

    // A = tangent with X axis: (4,0)
    pts[2] = 4.0; pts[3] = 0.0;

    // P = (1, sqrt(3))
    pts[4] = 1.0; pts[5] = sqrt(3.0);
```

C-Code

```
    // Compute direction vectors:
     // Tangent at P is x + sqrt(3)y = 4
     // Let y = t, x = 4 - sqrt(3)t; giving (x, y) = (4-sqrt(3)t,
         t)
     // Choose t1 = 0 for intersection A, and t2 = 2 for below
         intersection B
     // B = (4 - sqrt(3)*2, 2) = (4 - 2*1.73205, 2) ~ (0.5359, 2)
     pts[6] = 4.0 - sqrt(3.0)*2.0;
     pts[7] = 2.0;

     // Normal: line through O and P, param t, (t, t*sqrt(3))
     // With x axis, y=0 => t=0 (origin)
     // With y axis, x=0 => y=0 (again origin), for plotting
         purpose get a point far from origin t=2
     pts[8] = 2.0;
     pts[9] = 2.0*sqrt(3.0);
 }
```

Python-Code

```python
# Code by GVV Sharma
# Modified for Problem Solution
# Released under GNU GPL
# Calculating area enclosed between curves
# call.py
import ctypes
import numpy as np

def get_points():
    lib = ctypes.CDLL('./code.so')
    arr = (ctypes.c_double * 10)()
    lib.get_points(arr)
    return np.array(list(arr))

if __name__ == "__main__":
    pts = get_points()
    print(pts.reshape(-1, 2))
```

Python-Code

```
# plot.py
import numpy as np
import matplotlib.pyplot as plt
from call import get_points

pts = get_points()
O = pts[0:2]
A = pts[2:4] # tangent intersection (4, 0)
P = pts[4:6] # (1, sqrt(3))
T = pts[6:8] # other tangent direction point
N = pts[8:10] # other normal direction point

# Draw circle
theta = np.linspace(0, 2*np.pi, 400)
xc = 2 * np.cos(theta)
yc = 2 * np.sin(theta)
```

Python-Code

```
plt.figure(figsize=(7,7))
plt.plot(xc, yc, 'k--', label='Circle $x^2 + y^2 = 4$')

# Plot the tangent (from A towards T)
tan_x = [A[0], T[0]]
tan_y = [A[1], T[1]]
plt.plot(tan_x, tan_y, 'tab:red', label="Tangent at $P$")

# Plot the normal (from O towards N)
norm_x = [O[0], N[0]]
norm_y = [O[1], N[1]]
plt.plot(norm_x, norm_y, 'tab:green', label="Normal at $P$")

# Draw triangle OAP
tri_x = [O[0], A[0], P[0], O[0]]
tri_y = [O[1], A[1], P[1], O[1]]
plt.plot(tri_x, tri_y, 'b-', alpha=0.7, label='Triangle OAP')
```

Python-Code

```
# Points
plt.scatter([O[0], A[0], P[0]], [O[1], A[1], P[1]], color='purple', zorder=5)
plt.text(O[0], O[1]-0.2, "O", fontsize=12)
plt.text(A[0]+0.1, A[1]-0.1, "A", fontsize=12)
plt.text(P[0]+0.1, P[1]+0.1, "P", fontsize=12)

plt.xlim(-1, 5)
plt.ylim(-1, 5)
plt.gca().set_aspect('equal')
plt.grid(True)
plt.legend()
plt.title("Triangle formed by X axis, tangent, normal to $x^2 + y^2 = 4$\nat $P(1,\\sqrt{3})$")
plt.show()
```